

Haokai Ding

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EDUCATION

Shenzhen Technology University (SZTU)

Sept 2022 –Jul 2026

B.Eng. in Electronic Science & Technology; GPA: **84.6/100**, Rank: **12/176**

Research focus: LIBS spectroscopy; robotics systems.

Tsinghua University —Shenzhen X-Institute

Sept 2023 –Present

Tsien Excellence Engineering Program (TEEP), Joint Training (Underactuated Robotics)

PUBLICATIONS

An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping

2025

H. Ding[†], Y. Chen[†], D. Jia, W. Zhang^{*}

ICIA 2025. [Accepted, EI]

A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping

2025

H. Ding[†], W. Zhang^{*}

IROS 2025. [Accepted, EI]

Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping

2025

H. Ding[†], Z. Chen, T. Yang^{*}, W. Zhang^{*}

CASE 2025. [Published, EI]

Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp

2025

H. Ding[†], J. Fan, Z. Zhu, Y. Zhao, K. Chen^{*}, W. Zhang^{*}

ARM 2025. [Published, EI]

Rapid Analysis of Heavy Metal Element Adsorption by SCG Based on LIBS Technology

2025

H. Xie, L. You, X. Fang, L. Li, H. Ding, Z. Zhou, G. Zhang, D. Zhang

E3S Web Conf., 615 (2025) 02010. [EI]

Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries

2025

Y. Liu[†], H. Ding[†], H. Chen, H. Gao, J. Yu, F. Mo, N. Wang^{*}

Polymers 17(15):2136, 2025. [SCI, JCR Q1]

Legend: [†] Co-first author; ^{*} Corresponding author

RESEARCH & PROJECTS

Underactuated Robotic Gripper Research —Shenzhen X-Institute, Tsinghua University

2023 –Present

Designed and tested underactuated robotic grippers with semi-Peaucellier linkage.

Focus: idle-stroke / delay-triggering mechanisms; validated prototypes via simulation and experiments.

LIBS Spectroscopy for Heavy Metal Analysis —SZTU

2022 –2023

Built and optimized LIBS system for rapid adsorption analysis.

Focus: improved calibration pipeline and enhanced acquisition accuracy.

HONORS & AWARDS

ASME Student Mechanism & Robot Design Competition —Finalist International student competition finalist for mechanism and robot design.	2025
President’ s Award, SZTU Highest student honor at Shenzhen Technology University.	2024
China “Internet+” Innovation & Entrepreneurship (Guangdong) —Gold Prize Provincial-level first prize for innovation project.	2023
“Challenge Cup” (Guangdong Division) —Second Prize Provincial student innovation competition award.	2023
Chen Hsong Scholarship Dean’ s / Academic Excellence Scholarship, SZTU.	2023

SKILLS

Programming: C/C++, Python, MATLAB
Tools: SolidWorks, LTSpice, Arduino, Linux
Languages: Chinese (native), English (fluent)